

Yang-Jun Joo, Ph.D.

Data Scientist | Quantitative Risk Assessment

Autonomous Vehicle Safety | Statistical & Probabilistic Modeling | Large-Scale Driving Data

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Professional Summary: Quantitative risk assessment data scientist with 8+ years of relevant research and applied project experience, including 3+ years of post-Ph.D. experience, developing probabilistic safety metrics, uncertainty-aware models, and automated risk-assessment tooling for autonomous-vehicle and transportation systems. Author of 15 published or accepted papers and PI on two competitively funded safety projects, with experience in Python, SQL, large-scale driving data, and human-behavior experiments.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, pandas, NumPy, scikit-learn, PyTorch, Git, Linux, ETL and analytics pipelines.
Risk & Statistics:	Probabilistic modeling, Bayesian networks, reliability analysis, surrogate safety measures, regression, treatment-effect analysis, uncertainty quantification, conformal prediction.
Safety & Mobility:	Safety-performance metrics, AV safety evaluation, naturalistic driving video, connected-vehicle and trajectory data, traffic-conflict analysis, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar	University of Central Florida	Mar 2024 – Present
- Led FDOT quantitative safety workstreams with cross-functional agency and technical partners, defining and validating safety-performance metrics using high-resolution traffic-controller, connected-vehicle, and naturalistic driving data. - Built Python and SQL pipelines for multi-terabyte traffic and connected-vehicle data across 1,050 arterial links and four Florida freeways, automating data cleaning, risk-metric generation, validation, and reporting. - Developed uncertainty-aware evaluation methods for autonomous-driving vision-language models using conformal prediction, identifying context-specific calibration failures in safety-critical driving-scene question answering. - Designed and analyzed a controlled study of 926 participants, 265 events, and 1,325 video clips; used regression and heterogeneous treatment-effect analysis to quantify effects of automation, severity, and participant characteristics on perceived risk.		
Postdoctoral Researcher	Seoul National University	Mar 2023 – Feb 2024
- Led research on real-time traffic safety facility information integration, developing database management architectures and safety operational criteria for the Korean National Police Agency (KNPA). - Formulated cooperative traffic control strategies and system models to optimize real-time traffic safety and operations in mixed autonomy connected vehicle environments.		
Graduate Research Assistant	Seoul National University	Mar 2018 – Feb 2023
- Developed probabilistic and reliability-based risk-assessment models for autonomous-vehicle lane changes, individual-driver crash risk, and proactive collision avoidance using Bayesian networks, surrogate safety measures, and risk-field methods. - Secured and led a KRW 96.1M Global Ph.D. Fellowship, producing three first-author peer-reviewed publications.		

SELECTED PUBLICATIONS

- Joo et al. (2023), "A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory," *Analytic Methods in Accident Research*, 37, 100251.
- Joo et al. (2022), "A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment of Individual Driver with Traffic Violation and Crash Records," *Accident Analysis & Prevention*, 174, 106721.
- Joo et al. (2026), "Context-Aware Conformal Prediction for VLM-Based Driving Scene QA," *Forty-Third International Conference on Machine Learning (ICML) Workshop EMM-QA*.

EDUCATION

Ph.D., Civil & Environmental Engineering (Transportation), Seoul National University, 2023

